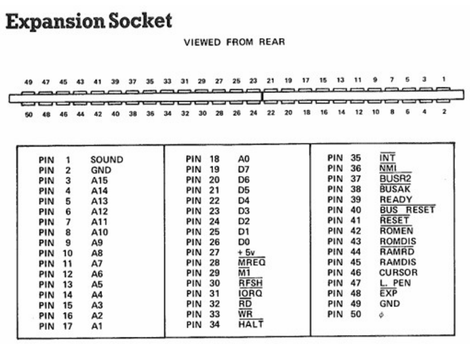
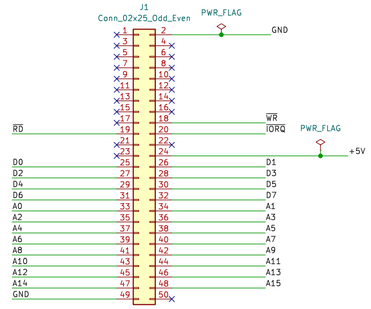
Amstrad CPC MX4 Connector

Female sockets on connector have pins reversed by orientation on the board so keep the same logical connections i.e. Pin 2 is Pin 49 and pin 3 is pin 48



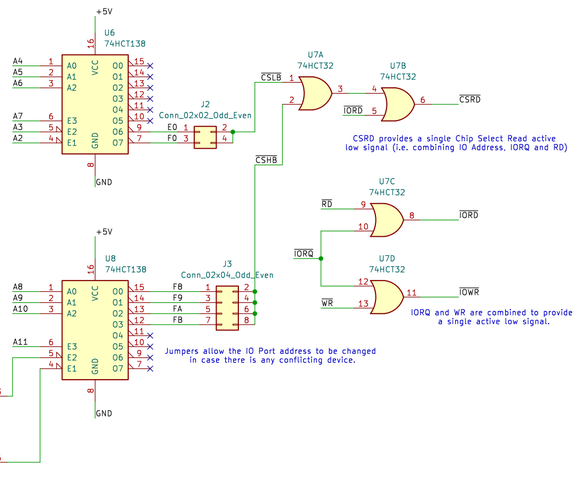


IO Read Address Decode and IO Write Signal

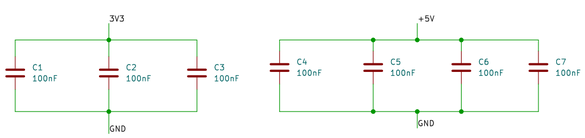
138 Decoders provide a low and high address byte selection signal (i.e. FBE0).
This is used to provide single active low signal for a port read when combined with IORQ+RD.

ADDRESS LINES							OUTPUTS								HEX ADDRESS
7	3	2	6	5	4										
INPUTS															
ENABLE			SELECT												
G1	/G2B	/G2A	C	B	A		Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	
X	1	X	X	X	X		1	1	1	1	1	1	1	1	
X	X	1	X	X	X		1	1	1	1	1	1	1	1	
0	X	X	X	X	X		1	1	1	1	1	1	1	1	
1	0	0	0	0	0		0	1	1	1	1	1	1	1	80
1	0	0	0	0	1		1	0	1	1	1	1	1	1	90
1	0	0	0	1	0		1	1	0	1	1	1	1	1	A0
1	0	0	0	1	1		1	1	1	0	1	1	1	1	B0
1	0	0	1	0	0		1	1	1	1	0	1	1	1	C0
1	0	0	1	0	1		1	1	1	1	1	0	1	1	D0
1	0	0	1	1	0		1	1	1	1	1	1	0	1	E0
1	0	0	1	1	1		1	1	1	1	1	1	1	0	F0
PINS															
6	5	4	3	2	1		15	14	13	12	11	10	9	7	

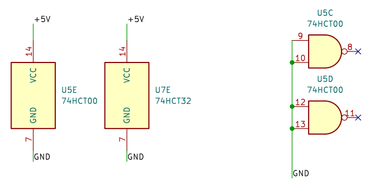
ADDRESS LINES							OUTPUTS								HEX ADDRESS
11/(14&15)	/(12&13)	10	9	8											
INPUTS															
ENABLE			SELECT												
G1	/G2B	/G2A	C	B	A		Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	
X	1	X	X	X	X		1	1	1	1	1	1	1	1	
X	X	1	X	X	X		1	1	1	1	1	1	1	1	
0	X	X	X	X	X		1	1	1	1	1	1	1	1	
1	0	0	0	0	0		0	1	1	1	1	1	1	1	F800
1	0	0	0	0	1		1	0	1	1	1	1	1	1	F900
1	0	0	0	1	0		1	1	0	1	1	1	1	1	FA00
1	0	0	0	1	1		1	1	1	0	1	1	1	1	FB00
1	0	0	1	0	0		1	1	1	1	0	1	1	1	FC00
1	0	0	1	0	1		1	1	1	1	1	0	1	1	FD00
1	0	0	1	1	0		1	1	1	1	1	1	0	1	FE00
1	0	0	1	1	1		1	1	1	1	1	1	1	0	FF00
PINS															
6	5	4	3	2	1		15	14	13	12	11	10	9	7	



POWER DECOUPLING



UNUSED GATES AND DEVICE POWER



USEFUL LINKS

CPC Wiki Amstrad CPC Default IO Ports
CPC Wiki Amstrad CPC IO Port Summary
CPC Wiki Amstrad CPC Expansion Port

Bus Transcivers

Bus Transceiver provide voltage level shift from 5V to 3V3 and Bus isolation

CONTROL (CTRL)

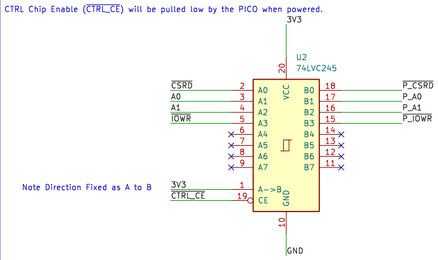
This transceiver buffers the control signals associated with IO Reads and Writes from the CPC.

Chip Select Read ($\overline{CSRD}$) is active low when CPC is reading the Pico simulated IO port i.e. FBE0. The Pico will use this as a trigger to place the next data byte on the data lines and change direction of the Data Transcievr.

A0 and A1 have also been provided as an optional address selection and may allocate these to return the CRTC, GA or PIO data but not decided yet.

$\overline{IOWR}$ active is low when the CPC is executing any IO Write, which will include the CRTC, GA or PIO. The PICO will use this as a trigger to log the data.

CTRL Chip Enable ($\overline{CTRL_CE}$) will be pulled low by the PICO when powered.



Address High Byte (ADHB)

This transceiver buffers the high byte of the IO Write from the CPC. Only the High Byte of the Address is used on the Amstrad CPC to select the CRTC, GA and PIO.

Note on the Amstrad CPC selection of the CRTC, GA and PIO is made with a single active low address line (i.e. B14 low selects the CRTC)

Address High Byte Chip Enable ($\overline{ADHB_CE}$) will be pulled low by the PICO when powered.

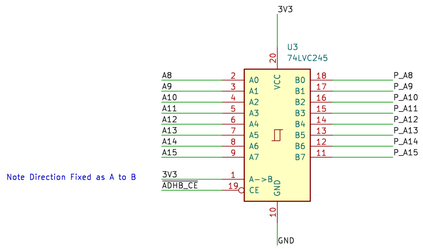
I/O Port Allocation Rules

Components use partial address decoding. It is therefore possible to send the same value to different components **simultaneously** by carefully using custom I/O ports.

Hardware device	Read/Write	Port bits															
		b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Gate Array	Write only	0	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PAL (RAM configuration)	Write only	0	*	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CRTC	Read/Write	-	0	-	-	-	-	r1	r0	-	-	-	-	-	-	-	-
Upper ROM select	Write only	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-	-
Printer Port	Write only	-	-	-	0	-	-	-	-	-	-	-	-	-	-	-	-
8255 PPI	Read/Write	-	-	-	-	0	-	r1	r0	-	-	-	-	-	-	-	-
Expansion Peripherals	Read/Write	-	-	-	-	-	0	x	x	x	x	x	x	x	x	x	x

Legend:

- "-" means this bit is ignored.
- "0" means the bit must be set to "0" for the hardware device to respond.
- "1" means the bit must be set to "1" for the hardware device to respond.
- "r1" and "r0" mean a bit used to define a register
- "*" Bit14 of the PAL port must be at 1 on CPCs equipped with CRTCs 0/1/2. It can be at 0 or 1 on CRTCs 3/4. For compatibility reasons, it is strongly advised to always set bit14 to 1 to select PAL.

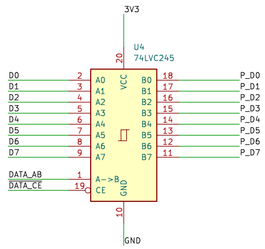


DATA

This transceiver buffers the high byte of the Data lines to and from the CPC.

The direction of data will be controlled by the Pico using Data Chip Enable (DATA_AB). By default this will be A to B and only on a IO Port Read will the Pico change this to B to A.

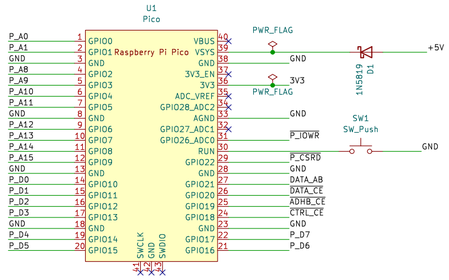
Data Chip Enable ($\overline{DATA_CE}$) will be pulled low by the PICO when powered.



PICO

One PIO will wait on $\overline{IORW}$ and when triggered place data in RX FIFO. DMA will move this into memory of CPU to decode and place in memory associated with IO Address.

One PIO will wait on $\overline{CSRD}$ and when triggered place data in TX FIFO. DMA will provide data into PIO. Not figured this bit out yet fully.



V1.1 Changed GPIO to make D0–D7 consecutive

Sheet: /
File: Amstrad-CPC-IO-Logger.kicad_sch

Title: RABS Amstrad CPC IO Data Logger

Size: A2	Date: 2026-02-15	Rev: V1.1
KiCad E.D.A. 9.0.6		Id: 1/1